

LECTURE 13



Slide 1 & 5: Vaccine Strategies Overview

These slides introduce the basic idea of vaccination and contrast two main approaches: using a **whole virus/bacterium** versus using **parts that trigger the immune system**.

- **The Basic Idea:** The fundamental principle of vaccination is to trigger a "**mild infection to expose your immune system to the pathogen**" to build immune memory¹¹¹¹.
- **Strategy 1: Using a Whole Virus or Bacterium:**
 - The pathogen must be **attenuated** (weakened) so it doesn't cause severe disease²²²².
 - A risk of this method is that the pathogen might "re-learn what cells it likes to infect" and potentially cause unintended damage³³³³.
 - This type of vaccine **must be grown in large quantities** in cells, tissues, or whole embryos (like chick embryos)⁴⁴⁴⁴.
- **Strategy 2: Using Parts or Genetic Material:** This approach focuses on using only the components (like proteins) or the **genetic material** (like mRNA) that elicit an immune response, rather than the whole, replicating pathogen⁵⁵⁵⁵.



Slides 2, 3, & 4: Recombinant DNA Technology for Vaccine Production (Hepatitis B Virus - HBV)

These slides detail the replication cycle of the Hepatitis B Virus (HBV) in a liver cell (hepatocyte) and explain how this process led to the development of a **recombinant vaccine** using biotechnology.

- **HBV Replication Cycle:** The HBV **Virion** enters the **Hepatocyte** by binding to receptors like the Heparan sulfate Na^+ bile acid proteoglycan symporter⁶⁶⁶⁶⁶⁶⁶⁶. The virus releases its partially double-stranded relaxed circular DNA (**rcDNA**) into the nucleus⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷⁷.
 - Inside the nucleus, the rcDNA is converted to a stable, fully double-stranded form known as **cccDNA** (covalently closed circular DNA)⁸⁸⁸⁸⁸⁸⁸⁸.
 - The cccDNA undergoes **Transcription** to produce various mRNA transcripts, including one for the **Hepatitis B Surface Antigen (HBsAg)**⁹⁹⁹⁹⁹⁹⁹⁹.

- [illegible]

 Slide 6: Concept of an mRNA Vaccine

This slide illustrates the mechanism of a newer vaccine type using messenger RNA (mRNA).

- **Delivery:** The **Spike protein mRNA** is packaged in **liposomes or other nano-carriers** (the **Vaccine**) to protect it and deliver it into the cell¹⁹.
- **Translation:** Once inside the **Cytoplasm** of the **Cell**, the mRNA does **not enter the nucleus**²⁰. It is immediately taken up by a **Ribosome**²¹.
- **Protein Synthesis:** The Ribosome translates the mRNA sequence into an **Amino acid chain**²², which folds to become the target protein, such as the **Spike protein**²³.

- **Immune Response:** The Spike protein is displayed by the cell, triggering the body to produce **Antibodies and other immune cells**²⁴.
- **Degradation:** The mRNA is temporary and is soon broken down (**Degraded mRNA**)²⁵.



Slide 7: Infectious Agents - Biomolecules

This slide introduces infectious agents that are not viruses or bacteria, consisting of only a biomolecule.

- **Viroids:**
 - These are **circular RNAs of ~ 100 bases**²⁶.
 - They primarily **infect plants**²⁷.
 - Crucially, they **do not encode for proteins**²⁸.
 - The image shows a plant infected with **Chrysanthemum Stunt Viroid (CSVd)**, which results in stunted growth and smaller flowers compared to a healthy plant²⁹²⁹²⁹²⁹.
- **Prions:**
 - These are **infectious proteins**³⁰.
 - They cause **degenerative brain diseases in animals** (e.g., Mad Cow Disease)³¹.
 - Prions are **mis-folded versions of normal proteins**³²³².
 - When an **Original prion** contacts a **Normal protein**, it **triggers a chain reaction**, converting the normal version into a new prion³³³³³³³³. These new prions then form **Aggregates of prions**³⁴.



Slides 8, 9, 10, & 11: The CRISPR-Cas System (Bacterial 'Vaccination')

These slides focus on the **CRISPR-Cas** system, the adaptive immune system of bacteria that protects them from viruses.

- **Background:** Organisms have evolved strategies to prevent or overcome viral infections³⁵. Complex organisms use a complex **immune system**³⁶. Bacteria, which are simpler organisms, use the CRISPR-Cas system to defend against **Bacteriophages**³⁷.
 - **CRISPR** stands for **C**lusters of **R**egularly **I**nter**S**paced **P**alindromic **R**epeats³⁸³⁸³⁸³⁸³⁸³⁸³⁸³⁸.
- **Mechanism (How Bacteria "Vaccinate" Themselves):**
 - **Phase I: Acquisition of CRISPR spacer (Memory)**³⁹³⁹³⁹³⁹³⁹³⁹³⁹³⁹
 - Upon initial infection by a **Bacteriophage**, the bacteria acquires viral genome sequences (**Protospacer**) and inserts them into its chromosome as **Spacer** sequences⁴⁰⁴⁰⁴⁰⁴⁰.
 - The Spacers, which are the "**memory**" of the encounter, are separated by **bacterial DNA Repeats**⁴¹. This occurs during an unsuccessful infection⁴².
 - The Protospacer is next to a **PAM** sequence (**Protospacer Adjacent Motif**) on the viral DNA⁴³⁴³⁴³⁴³⁴³⁴³⁴³⁴³.
 - **Phase II: Transcription and Formation of Cas-RNA Complexes (Activation)**⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴⁴
 - **CRISPR Associated genes (\$cas\$ genes)** convert the **spacer** and **repeat** sequences into a long RNA transcript (**Pre-crRNA**)⁴⁵.
 - This transcript is processed into short **crRNA** molecules, which become the **Guide RNA**⁴⁶⁴⁶⁴⁶⁴⁶⁴⁶⁴⁶⁴⁶⁴⁶.
 - \$cas\$ genes also produce the **Cas protein**, which is a nuclease (cuts DNA/RNA)⁴⁷.
 - The **Guide RNA** combines with the **Cas protein** to form a **Surveillance complex**⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸⁴⁸.
 - **Phase III: Activation of DNA Cutting Mechanism (Defense)**⁴⁹⁴⁹⁴⁹⁴⁹⁴⁹⁴⁹⁴⁹⁴⁹
 - Upon **Reinfection**⁵⁰⁵⁰⁵⁰⁵⁰⁵⁰⁵⁰⁵⁰⁵⁰, the complex recognizes the **Viral DNA**⁵¹⁵¹⁵¹⁵¹.

- The complex targets the viral nucleic acid due to **base complementarity** between the Guide RNA sequence and the viral DNA sequence⁵².
- The Cas protein then **cuts the invading Viral DNA**, neutralizing the infection⁵³.

Slides 12, 13, 14, & 15: CRISPR for Gene Editing (The "Bulls Eye" Tool)

These slides discuss the significance of the CRISPR-Cas system as a revolutionary **gene-editing tool**.

- **Discovery and Recognition:**
 - **Francisco Mojica** originally identified the spacer DNAs from viral genomes in bacteria and archaea and **coined the term CRISPR** around 2005⁵⁴⁵⁴⁵⁴⁵⁴.
 - **Emmanuelle Charpentier** and **Jennifer Doudna** were awarded the **2020 Nobel Prize in Chemistry** for their work detailing the mode of action of CRISPR⁵⁵⁵⁵⁵⁵⁵⁵.
- **The Significance of CRISPR (The "Bulls Eye"):**
 - Before 2012, methods for causing DNA mutations (e.g., chemicals, radiation) were random; if DNA were a dartboard, you could ensure the dart would hit the board, but not a specific target⁵⁶.
 - **With CRISPR, we can hit the bulls eye**—it allows for targeted, specific gene editing⁵⁷.
- **What Makes CRISPR Specific?**
 - **Specificity** is achieved through **sequence complementarity between the RNA and DNA**⁵⁸.
 - If you know the sequence of the DNA, you can predict the sequence of the RNA needed⁵⁹⁵⁹⁵⁹⁵⁹.
 - That specific RNA (**Guide RNA**) is made in a lab (*in vitro*) and mixed with the Cas9 protein⁶⁰.
 - This RNA-Cas9 complex "knows where to go" because the RNA sequence is designed to target only one specific DNA sequence⁶¹⁶¹⁶¹⁶¹.
- **Gene Editing with Custom CRISPR-Cas9:**

- Genome editing requires a single guide (**sgRNA**) to direct the **Cas9 endonuclease** to a specific genomic region⁶².
- The Cas9 complex **searches the DNA for PAM binding sites**⁶³.
- If the **PAM** is found, Cas9 binds, forming an **R-loop** with the DNA⁶⁴. The PAM sequence is required next to the site of cleavage⁶⁵.
- Cas9 then **generates a double-stranded DNA break**⁶⁶⁶⁶⁶⁶⁶⁶.
- The cut DNA is then left for repair by other cellular enzymes, which can be manipulated for gene editing purposes⁶⁷⁶⁷⁶⁷⁶⁷.

1. Hepatitis B Virus (HBV) Replication Cycle

This diagram illustrates the life cycle of the Hepatitis B Virus within a liver cell (**Hepatocyte**). This process highlights the production of viral components, which is leveraged for recombinant vaccine technology.

- **Entry and Uncoating:** The **Virion** enters the **Hepatocyte** via receptors like the Heparan sulfate Na^+ bile acid proteoglycan symporter, a process labeled **Entry**¹¹¹¹. The partially double-stranded **rcDNA** (relaxed circular DNA) is then released²².
- **Genome Establishment:** The rcDNA is transported to the nucleus and converted into **cccDNA** (covalently closed circular DNA), the stable transcriptional template³.
- **Transcription:** The cccDNA undergoes **Transcription** to produce various viral RNA transcripts, including one for the **pgRNA** (pre-genomic RNA) and one for the **HBsAg** (Hepatitis B Surface Antigen)⁴⁴⁴⁴.
- **Protein Synthesis & Processing:** HBsAg is encoded by the gene S⁵. This protein is translated and processed through the **ER** (Endoplasmic Reticulum) and **Golgi apparatus**⁶.
- **Assembly and Reverse Transcription:** The pgRNA is packaged into newly assembled capsids (**Translation & capsid assembly**)⁷. Inside the capsid, the pgRNA is reverse-transcribed into **ssDNA** (single-stranded DNA) and then back to **rcDNA**⁸⁸⁸⁸.
- **Release:** The new virions mature through **Morphogenesis** and are processed by a **Multivesicular body** before being released⁹.

- **Subviral Particles:** The infection results in the **overproduction of 22nm particles** made exclusively of **HBsAg**¹⁰. These are non-infectious **HBs subviral particles** that serve as the immunogenic component of the vaccine¹¹.

2. Concept of an mRNA Vaccine

This diagram details the mechanism of a modern mRNA vaccine, showing how genetic material is used to elicit a protective immune response without causing infection.

- **Delivery:** The **Vaccine** contains **Spike protein mRNA** packaged in **liposomes or other nano-carriers**¹². The vaccine enters the **Cell** and releases the mRNA into the **Cytoplasm**¹³¹³¹³¹³. The mRNA does **not enter the Nucleus**¹⁴.
- **Translation (Protein Production):** A **Ribosome** binds to the Spike protein mRNA and **translates** the genetic code into an **Amino acid chain**¹⁵.
- **Antigen Presentation:** The amino acid chain folds into the **Spike protein**¹⁶. This protein acts as the antigen, which is displayed by the cell.
- **Immune Response:** The host immune system recognizes the Spike protein as foreign, leading to the activation of **Antibodies and other immune cells**¹⁷. This process establishes immune memory.
- **Degradation:** The instructional mRNA is temporary and is broken down by cellular enzymes (**Degraded mRNA**) after it has been used¹⁸.

3. CRISPR-Cas System (Bacterial Defense)

This figure illustrates the adaptive immune system used by a **Microbe** (bacterium or archaeon) to defend against a **Bacteriophage (virus)**.

I. Acquisition of CRISPR Spacer (Memory)

- A **Bacteriophage** injects its **Viral DNA** into the microbe.
- A segment of the Viral DNA (**Protospacer (viral gene)**) and an adjacent **PAM** (**Protospacer Adjacent Motif**) is cleaved¹⁹¹⁹¹⁹¹⁹.
- This viral segment is inserted into the **Microbial DNA** as a **New spacer**, separated by **Repeats**²⁰²⁰²⁰²⁰. This spacer is the microbe's "memory" of the infection²¹.

II. Transcription and formation of Cas-RNA complexes (Activation)

- The entire CRISPR region (repeats and spacers) is transcribed into a long **Pre-crRNA transcripts**²².
- This pre-crRNA is processed into shorter **crRNA** molecules, which serve as the **Guide RNA**²³.
- The **Guide RNA** associates with a **Cas protein** (a nuclease) to form a **Surveillance complex**²⁴²⁴²⁴²⁴.

III. Activation of DNA cutting mechanism (Defense)

- Upon **Reinfection**, the Surveillance complex uses the **Guide RNA** to locate the complementary **Same viral gene** sequence in the invading **Viral DNA**²⁵.
- The **Cas protein** then cuts the viral DNA, neutralizing the threat²⁶.

4. Custom-Designed CRISPR-Cas9 System (Gene Editing)

This figure demonstrates the use of the engineered CRISPR-Cas9 system for precise gene editing within a **Eukaryotic cell**.

- **System Components:** The system consists of a custom-designed **Guide RNA** bound to the **Cas9** protein (an endonuclease)²⁷. The entire complex is delivered into the Eukaryotic cell²⁸.

Targeted DNA Cleavage for Editing

- I. Search for PAM: The Cas9-Guide RNA complex **searches the DNA for PAM binding sites**²⁹. The **PAM (Protospacer Adjacent Motif)** is required next to the target site³⁰.
- II. Binding and R-Loop Formation: **If PAM is found**, Cas9 binds, and the Guide RNA base-pairs with the complementary strand of the DNA, forming an **R-loop**³¹³¹³¹³¹.
This base complementarity is what makes the system specific³².
- III. Double-Stranded Break: Once bound, **Cas9 generates a double-stranded DNA break** at the target site³³.

- IV. Repair: The cut DNA is then **left for repair by other enzymes**³⁴. Scientists exploit these repair mechanisms to introduce specific changes (edits) to the genome.